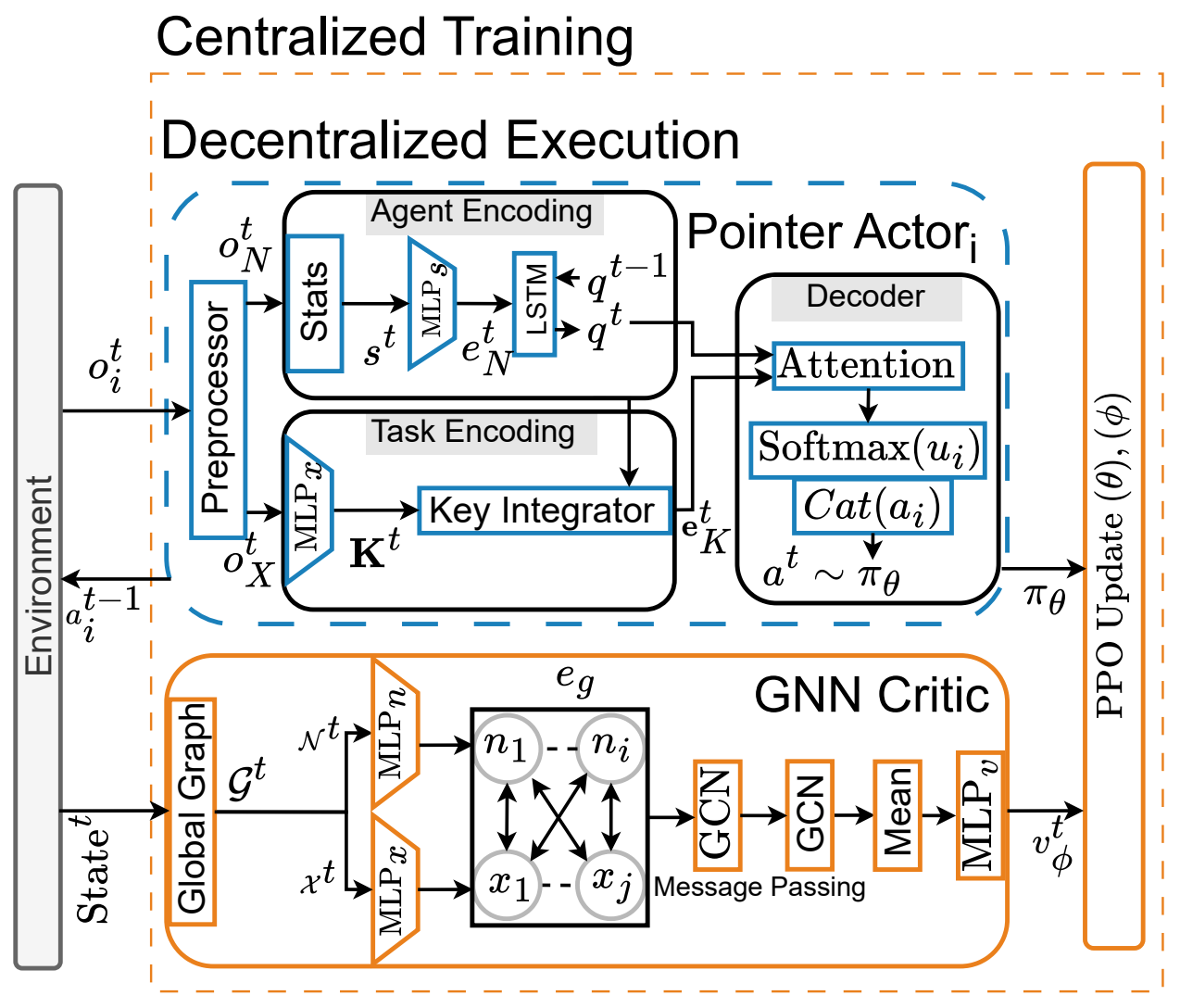

Centralized Training
Decentralized Execution
Environment
o_i^t
a_i^{t-1}
Preprocessor
o_N^t
o_X^t
Agent Encoding
Stats
s^t
MLP s
e_N^t
LSTM
q^{t-1}
q^t
Task Encoding
MLP x
$\mathbf{K}^t$
Key Integrator
e_K^t
Pointer Actor$_i$
Decoder
Attention
Softmax(u_i)
$Cat(a_i)$
$a^t \sim \pi_\theta$
π_θ
PPO Update $(\theta), (\phi)$
State^t
Global Graph
$\mathcal{G}^t$
$\mathcal{N}^t$
$\mathcal{X}^t$
MLP n
MLP x
e_g
GNN Critic
n_1
n_i
x_1
x_j
GCN
GCN
Mean
MLP v
Message Passing
v_ϕ^t